

1.

3.2.3 (2016 SP:04)

Which one of the following reactions does **not** represent the Brønsted-Lowry model?

- (a) $\text{HSO}_4^-(aq) + \text{H}_2\text{O}(\ell) \rightarrow \text{SO}_4^{2-}(aq) + \text{H}_3\text{O}^+(aq)$
- (b) $\text{CH}_3\text{COOH}(aq) + \text{NaOH}(aq) \rightarrow \text{NaCH}_3\text{COO}(aq) + \text{H}_2\text{O}(\ell)$
- (c) $\text{HCO}_3^-(aq) + \text{H}_2\text{O}(\ell) \rightarrow \text{H}_2\text{CO}_3(aq) + \text{OH}^-(aq)$
- (d) $\text{CaCO}_3(s) + 2 \text{HCl}(aq) \rightarrow \text{Ca}^{2+}(aq) + \text{CO}_2(g) + 2 \text{H}_2\text{O}(\ell)$

2.

3.2.8 (2016 SP:10)

Over the last 200 years, the pH of oceans has dropped from 8.2 to 8.1. A drop of 0.1 pH units represents an

- (a) approximate 20% increase in the concentration of hydrogen ions.
- (b) increase of the hydrogen ion concentration by a factor of 10.
- (c) approximate 20% increase in pH.
- (d) insignificant change in hydrogen ion concentration, due to the large volume of the ocean.

3.

3.2.11 (2016:07)

In an acid-base titration, which of the following is **least** likely to cause an error in the calculated concentration?

- (a) using a funnel in the burette and leaving it in the same place for each titration
- (b) measuring the volume at the bottom of the meniscus
- (c) each member of the experimental team taking turns to measure the burette
- (d) rinsing the burette with distilled water before the titration

4.

3.2.1 (2016:09)

How many moles of a diprotic acid would be required to neutralise 1 mole of sodium hydroxide?

- (a) 0.5
- (b) 1.0
- (c) 1.5
- (d) 2.0

5.

3.2.3 (2016:10)

Which one of the following represents a conjugate acid-base pair?

- (a) $\text{N}^{3-}/\text{CN}^-$
- (b) $\text{NH}_3/\text{NH}_2^-$
- (c) $\text{CH}_3\text{CH}_2\text{OH}/\text{CH}_3\text{CHO}$
- (d) $\text{H}_3\text{PO}_4/\text{PO}_4^{3-}$

6.

3.2.3 (2016:11)

Which of the following equations **best** represents the self-ionisation of water according to the Brønsted-Lowry model?

- (a) $\text{H}_2\text{O}(\ell) \rightleftharpoons \text{H}^+(\text{aq}) + \text{OH}^-(\text{aq})$
- (b) $\text{H}_2\text{O}(\ell) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{OH}^-(\text{aq})$
- (c) $2 \text{H}_2\text{O}(\ell) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + 2 \text{OH}^-(\text{aq})$
- (d) $2 \text{H}_2\text{O}(\ell) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{OH}^-(\text{aq})$

7.

3.2.10 (2016:12)

The **best** definition of the equivalence point in an acid-base titration is the point at which the

- (a) indicator changes colour.
- (b) volume of acid equals the volume of base.
- (c) pH of the solution is 7.
- (d) mole ratio of acid to base is equal to their stoichiometric ratio.

8.

3.2.2 (2017:01)

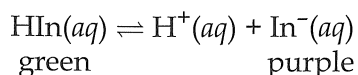
Which one of the following pairs contains a strong acid and a weak acid?

- (a) HCl and NaOH
- (b) MgCO_3 and CH_3COOH
- (c) NH_3 and KOH
- (d) HNO_3 and H_2CO_3

9.

3.2.9 (2017:08)

The indicator HIn is used in a titration between hydrochloric acid and magnesium hydroxide solutions. The following equation represents how the indicator works.



The indicator is added to 20.0 mL of magnesium hydroxide solution in a conical flask and the hydrochloric acid is added via a burette until the end point is observed. The acidic and basic solutions are of similar concentrations and the flask is swirled continuously as the acid is added.

Which one of the following statements describes the expected observations for the colour of the solution in the conical flask?

- (a) The solution starts green and turns purple after the addition of approximately 10 mL.
- (b) The solution starts green and turns purple after the addition of approximately 40 mL.
- (c) The solution starts purple and turns green after the addition of approximately 10 mL.
- (d) The solution starts purple and turns green after the addition of approximately 40 mL.

10.

3.2.9 (2018:01)

An acid-base indicator is red in acid, green in base and yellow in neutral solutions. The indicator was originally in sodium hydroxide solution and excess nitric acid was added dropwise. Which of the following shows the order of colour that would be shown by the indicator?

- (a) red only
- (b) yellow, green, red
- (c) green, yellow, red
- (d) yellow, red

11.

3.2.2 (2018:03)

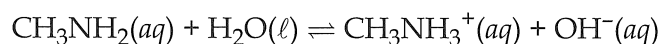
Which of the following solutions has the **greatest** electrical conductivity?

- (a) 0.010 mol L⁻¹ methanoic acid, HCOOH(aq) ($K_a = 1.8 \times 10^{-4}$)
- (b) 0.100 mol L⁻¹ methanoic acid, HCOOH(aq) ($K_a = 1.8 \times 10^{-4}$)
- (c) 0.010 mol L⁻¹ hypochlorous acid, HClO(aq) ($K_a = 3.5 \times 10^{-8}$)
- (d) 0.100 mol L⁻¹ hypochlorous acid, HClO(aq) ($K_a = 3.5 \times 10^{-8}$)

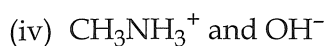
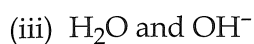
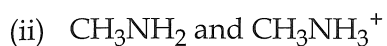
12.

3.2.4 (2018:08)

Consider the hydrolysis equation below.



Which of the following are conjugate acid-base pairs?



(a) ii only

(b) ii and iii only

(c) i and iv only

(d) i, ii, iii and iv

13.

3.1 (2018:18)

Which of the following sets of equations corresponds correctly to the acid-base theory of the chemist/s who proposed it?

	Chemist/s	Equations
(a)	Johannes Brønsted and Thomas Lowry Humphry Davy Svante Arrhenius	$\text{H}^+(aq) + \text{OH}^-(aq) \rightarrow \text{H}_2\text{O}(\ell)$ $\text{HNO}_3(aq) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{H}_3\text{O}^+(aq) + \text{NO}_3^-(aq)$ $\text{HC}_2\text{H}_3\text{O}_2(aq) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{C}_2\text{H}_3\text{O}_2^-(aq) + \text{H}_3\text{O}^+(aq)$
(b)	Johannes Brønsted and Thomas Lowry Humphry Davy Svante Arrhenius	$\text{HC}_2\text{H}_3\text{O}_2(aq) + \text{CH}_3\text{OH}(aq) \rightleftharpoons \text{CH}_3\text{OH}_2^+(aq) + \text{C}_2\text{H}_3\text{O}_2^-(aq)$ $\text{H}^+(aq) + \text{OH}^-(aq) \rightarrow \text{H}_2\text{O}(\ell)$ $\text{NH}_3(g) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{NH}_4^+(aq) + \text{OH}^-(aq)$
(c)	Johannes Brønsted and Thomas Lowry Humphry Davy Svante Arrhenius	$\text{HCl}(aq) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{H}_3\text{O}^+(aq) + \text{Cl}^-(aq)$ $\text{H}_3\text{O}^+(aq) + \text{OH}^-(aq) \rightarrow 2 \text{H}_2\text{O}(\ell)$ $\text{HNO}_3(aq) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{H}_3\text{O}^+(aq) + \text{NO}_3^-(aq)$
(d)	Johannes Brønsted and Thomas Lowry Humphry Davy Svante Arrhenius	$\text{NH}_3(aq) + \text{CH}_3\text{OH}(aq) \rightleftharpoons \text{CH}_3\text{O}^-(aq) + \text{NH}_4^+(aq)$ $2 \text{HCl}(aq) + \text{Mg}(s) \rightarrow \text{H}_2(g) + \text{Mg}^{2+}(aq) + 2 \text{Cl}^-(aq)$ $\text{NaOH}(s) \rightarrow \text{Na}^+(aq) + \text{OH}^-(aq)$

14.

3.2.9 (2019:01)

Boric acid, which is a weak acid, was titrated with standardised sodium hydroxide solution.

Which one of the indicators listed below would be the most suitable to use in this titration?

	Indicator	Range of colour change (pH)
(a)	thymol blue	1-3
(b)	bromocresol green	3.8-5.4
(c)	cresolphthalein	8-10
(d)	alizarin yellow	10-12

15.

3.2.2 (2019:08)

A distinguishing feature of strong acids is that they

- (a) produce high concentrations of hydronium ions (H_3O^+) in solution.
- (b) have high acidity constants.
- (c) contain loosely-held hydrogen ions (H^+) in solution.
- (d) ionise rather than dissociate in water.

16.

3.2.8 (2019:16)

Which one of the following statements about an aqueous solution with a pH less than zero at 25.0 °C is true?

- (a) Such a solution cannot exist at 25.0 °C.
- (b) There are no $\text{OH}^-(aq)$ ions present.
- (c) The concentration of $\text{H}^+(aq)$ ions is much greater than the concentration of $\text{OH}^-(aq)$ ions.
- (d) There are no $\text{H}^+(aq)$ ions present as they have formed water molecules through the process of neutralisation.

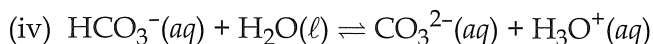
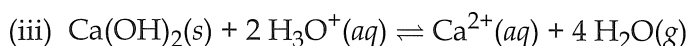
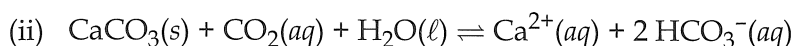
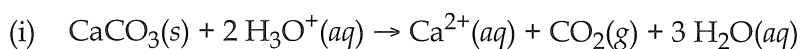
17.

3.2.4 (2019:21)

A group of Year 12 Chemistry students wanted to know whether increasing ocean acidity increases the rate at which sea shells, CaCO_3 , dissolve. They went to a beach to collect seawater and sea shells. In their school laboratory they crushed the sea shells and added 2.00 g of the resulting powder to five clean 250 mL beakers, each of which had been placed on top of its own electronic balance.

They split the seawater into five portions and bubbled carbon dioxide gas into four of the portions for different amounts of time. This gave the students 'natural' seawater plus four seawater samples of different pH. The various seawaters (150 mL portions) were then added to the beakers, with the weight of each beaker and its contents being recorded at timed intervals.

Which of the following reactions is/are likely to be occurring within the beakers during the investigation?

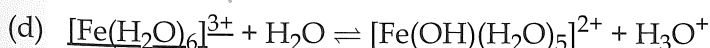
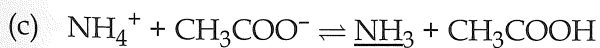
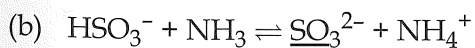
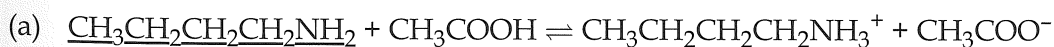


- (a) i and ii only
- (b) i, ii and iv only
- (c) iii only
- (d) i, ii, iii and iv

18.

3.1 (2019:24)

Which one of the following underlined species is acting as an acid?



19.

3.2.1 (2020:02)

Which of the following correctly classifies the given acids as monoprotic or polyprotic?

	Monoprotic	Polyprotic
(a)	HCl	CH ₃ COOH
(b)	CH ₃ CH ₂ COOH	H ₂ SO ₄
(c)	CH ₃ COOH	CH ₃ CH ₂ COOH
(d)	H ₂ SO ₄	HCl

20.

3.2.6 (2020:05)

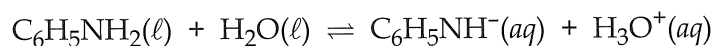
Which of the following statements about pure water are correct?

- (i) Pure water is a weak electrolyte that undergoes self-ionisation.
- (ii) The equilibrium constant for the ionisation of pure water at 25 °C is 1.00×10^{-14} .
- (iii) Pure water ionises completely at 25 °C, hence $[H^+] = [OH^-]$.
- (iv) The ionisation of pure water produces twice as many hydrogen ions as hydroxide ions.

- (a) i and ii only
- (b) ii and iii only
- (c) iii and iv only
- (d) i, ii, iii and iv

21.

3.2.3 (2020:15)

The reaction of aniline (C₆H₅NH₂) with water is an equilibrium process:

A conjugate acid-base pair in this process is

- (a) C₆H₅NH⁻ and H₂O
- (b) C₆H₅NH₂ and C₆H₅NH⁻
- (c) C₆H₅NH⁻ and H₃O⁺
- (d) H₃O⁺ and C₆H₅NH₂

22.

3.2.9 (2020:17)

Acid-base indicators

- (a) are oxidising or reducing agents.
- (b) change colour at a specific pH value.
- (c) are strong acids or bases.
- (d) are weak acids or bases.

23.

3.2.2 (2020:18)

A chemist prepares solutions of nitrous acid and hydrocyanic acid that have the same concentration.

The K_a values of these acids are:

- nitrous acid (HNO_2) is 4.6×10^{-4}
- hydrocyanic acid (HCN) is 6.17×10^{-10} .

Which of these two acids is the stronger and which has the higher pH?

	Stronger acid	Higher pH
(a)	nitrous acid	nitrous acid
(b)	nitrous acid	hydrocyanic acid
(c)	hydrocyanic acid	hydrocyanic acid
(d)	hydrocyanic acid	nitrous acid

24.

3.2.1 (2021:03)

Consider the acids listed in the following table.

Name	Formula	K_a (25 °C)
bromoacetic acid	CH_2BrCOOH	1.38×10^{-3}
dibromoacetic acid	CHBr_2COOH	3.31×10^{-2}
tribromoacetic acid	CBr_3COOH	1.91×10^{-1}

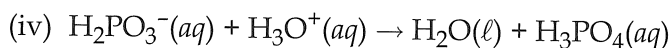
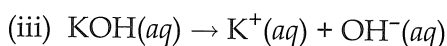
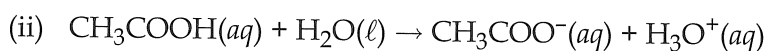
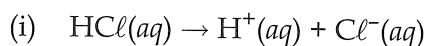
Which of the following identifies the strongest acid and classifies it correctly as monoprotic, diprotic or triprotic?

	Strongest acid	Classification
(a)	bromoacetic acid	monoprotic
(b)	dibromoacetic acid	diprotic
(c)	tribromoacetic acid	monoprotic
(d)	tribromoacetic aid	triprotic

25.

3.1 (2021:08)

Which of the following equation/s demonstrate/s the Arrhenius model of acids and bases?



(a) i, ii, iii and iv

(b) i only

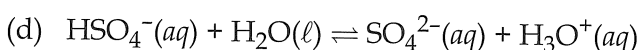
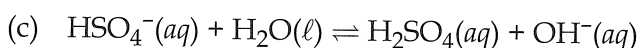
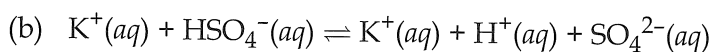
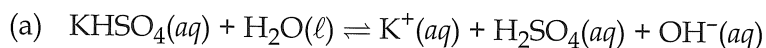
(c) ii and iii only

(d) i and iii only

26.

3.2.4 (2021:09)

The net ionic equation for the predominant hydrolysis reaction occurring in a 1.00 mol L^{-1} potassium hydrogensulfate solution is:



27.

3.2.8 (2021:14)

Consider the following pH values for a range of substances.

pH	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Substance	Battery acid	Stomach acid	Lemon juice	Orange juice	Beer	Black coffee	Milk	Blood	Sea water	Toothpaste	Laundry detergent	Bathroom cleaner	Hair straightener	Oven cleaner	Drain cleaner

Based on these pH values, which one of the following statements about the concentration of hydronium ions is true?

(a) It is 1000 times greater in sea water than in bathroom cleaner.

(b) It is twice as great in beer as it is in lemon juice.

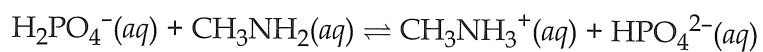
(c) It is three times greater in beer than in hair straightener.

(d) It is 1 000 000 times greater in hair straightener than in milk.

28.

3.2.3 (2021:21)

Identify a conjugate acid-base pair in the reaction represented by the following equation:



- (a) $\text{H}_2\text{PO}_4^-(aq)$ and $\text{CH}_3\text{NH}_2(aq)$
- (b) $\text{CH}_3\text{NH}_3^+(aq)$ and $\text{HPO}_4^{2-}(aq)$
- (c) $\text{H}_2\text{PO}_4^-(aq)$ and $\text{HPO}_4^{2-}(aq)$
- (d) $\text{H}_2\text{PO}_4^-(aq)$ and $\text{CH}_3\text{NH}_3^+(aq)$

Chapter 2: Acids and Bases

- 1.(2016 SP:04-d) The Brønsted-Lowry model is about accepting and losing protons. Answer 'd' is an acid/carbonate reaction. The answer is 'd'.
- 2.(2016 SP:10-a) Original pH was $8.2 = [\text{H}^+]$ of 6.31×10^{-9} and the new $[\text{H}^+] = 7.94 \times 10^{-9}$ so there was a 20% change. The answer is 'a'.
- 3.(2016:07-b) Leaving the funnel in place will add drops after the initial reading, having many people making readings will add random errors, rinsing with distilled water will dilute the titrant but measuring from the bottom of the meniscus for both the initial and final reading is not an error so the answer is 'b'.
- 4.(2016:09-a) A diprotic acid has two protons e.g. H_2SO_4 . As the mol ratio of $\text{NaOH}:\text{H}_2\text{SO}_4$ is 2:1 we need half the mol. The answer is 'a'.
- 5.(2016:10-b) A conjugate acid base pair must differ by only one hydrogen. So a and d are eliminated. Answer c is a redox reaction. The answer is 'b'.
- 6.(2016:11-d) The Brønsted-Lowry model deals with gain and loss of H^+ and uses H_2O and H_3O^+ so answer 'a' is wrong. Answers 'b' and 'c' are not balanced so the answer is 'd'.
- 7.(2016:12-d) The question seeks to clarify the difference between end point and equivalence point so answer 'a' is wrong. Answer 'b' is only right if the stoichiometric ratio is 1:1 and the concentrations are 1:1 or the stoichiometric ratio $\times 1/\text{concentration}$ is a constant. Answer 'c' is wrong because the pH is not 7 when weak acids or bases are titrated against strong ones. The answer is 'd'.
- 8.(2017:01-d) This question combines knowing of what an acid is and knowing the names of a very limited number of strong acids; including nitric, hydrochloric and sulfuric. The answer is 'd'.
- 9.(2017:08-d) The question reinforces the need to write an equation before starting a chemistry problem. The mol ratio is 2:1 so the volumes will be in the same ratio. You should also note the base was in the flask so the answer is 'd'.
- 10.(2018:01-c) Interesting fact that mixing red and green makes yellow. The solution began basic so the initial indicator state would be green and would, at equivalence, be yellow and then turn red in acid solution. The answer is 'c'.
- 11.(2018:03-b) The conductivity of a solution depends on the concentration of ions present. Thus, the acid with the largest K value and the highest concentration will have the most ions and conduct best. The answer is 'b'.
- 12.(2018:08-b) A conjugate pair differ by just one H. So only (ii) and (iii) meet this condition. The answer is 'b'.
- 13.(2018:18-d) Davy considered an acid to have replaceable hydrogen, while Arrhenius said acids produce H^+ in aqueous solution and bases produce OH^- in aqueous solution. Brønsted/Lowry considered acids as proton donors and bases as proton acceptors. The answer is 'd'.
- 14.(2019:01-c) When boric acid is neutralised by sodium hydroxide solution the borate ion will be formed. This ion will undergo hydrolysis and reform boric acid as it is a weak acid. Hydroxide ions will form, the $[\text{OH}^-] > [\text{H}^+]$ so the pH will be above 7. To find this equivalence point requires an indicator with an end point of 8–10 – the answer is cresolphthalein – the answer is 'c'.
- 15.(2019:08-b) It is important to realise answer (a) is not correct because that might refer to a high concentration of weak acid and similarly strong acids might be dilute and not produce large amounts of hydrogen ions. The answer is 'b'.
- 16.(2019:16-c) Commercial grade $\text{HCl}(\text{aq})$ can be purchased at 12 mol L^{-1} . This has a pH of -1.07 . The pH scale is designed for common, especially biological, systems and, in unusual instances, extends beyond 0 and 14. The answer is 'c'.
- 17.(2019:21-b) The experiment involves calcium carbonate and calcium hydrogencarbonate. There was no calcium hydroxide present. The answer is 'b'.
- 18.(2019:24-d) In three cases the underlined substance is acting as a Brønsted-Lowry base. The answer is 'd'.
- 19.(2020:02) A monoprotic acid has only one acidic hydrogen while a polyprotic acid has more than one. To be acidic a hydrogen must bond to an oxygen or atom of high electronegativity eg Cl in HCl but not C in CH_3^- . The answer is (b).
- 20.(2020:05) i) is correct ii) is correct
iii) is not true – $[\text{H}^+] = [\text{OH}^-]$ but the ionisation is not complete
iv) is not true – $[\text{H}^+] = [\text{OH}^-]$
Answer is (a).
- 21.(2020:15) A conjugate acid base pair are two molecules where the second is formed by the first losing an H^+ . The answer is (b).
- 22.(2020:17) Indicators change over a range of pH values so (b) is wrong. If they were strong they would not change colours. They are weak acids or bases that happen to demonstrate a colour change. The answer is (d).
- 23.(2020:18) Stronger means produces more product – the answer is nitrous acid. This is shown by a higher K_a value. Higher pH means produces fewer H^+ ions – the answer is hydrocyanic acid. The answer is (b).
- 24.(2021:03) The strongest acid will have the largest K_a value. The only acidic hydrogens are those connected to oxygen atoms. The answer is c.
- 25.(2021:08) Arrhenius's definition was restricted to aqueous solutions and involved the formation of H^+ and OH^- ions. The answer is d.
- 26.(2021:09) $\text{HSO}_4^-(\text{aq})$ is the conjugate of the very strong acid H_2SO_4 and so is, itself acidic. It will donate a proton. The answer is d.
- 27.(2021:14) This question is reminding us that the pH scale is logarithmic. Seawater and bathroom cleaner differ by 3 pH units = $10^3 = 1000$. The answer is a.
- 28.(2021:21) A conjugate acid-base pair differ by just one hydrogen. The answer is c.